

Program 10

Demonstrate Minikube Dashboard using CLI and Kubernetes Cluster

Project Structure

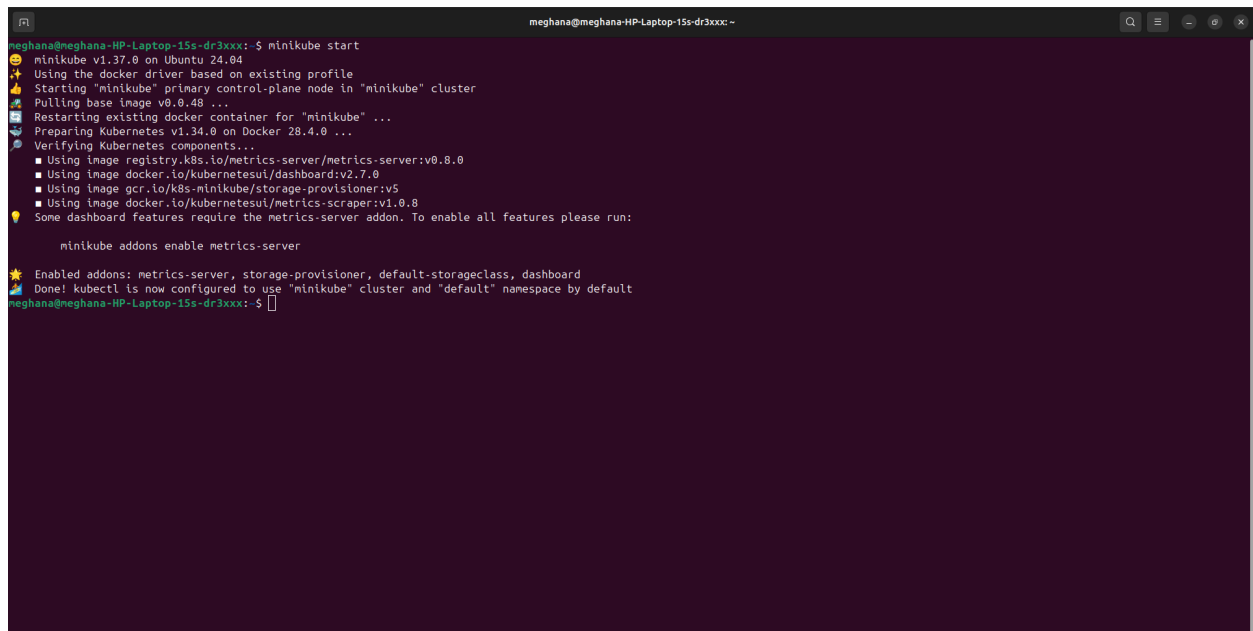
```
program_10/  
├── deployment.yaml  
└── services.yaml
```

Step 1: Create Project Directory

```
mkdir program_10  
cd program_10
```

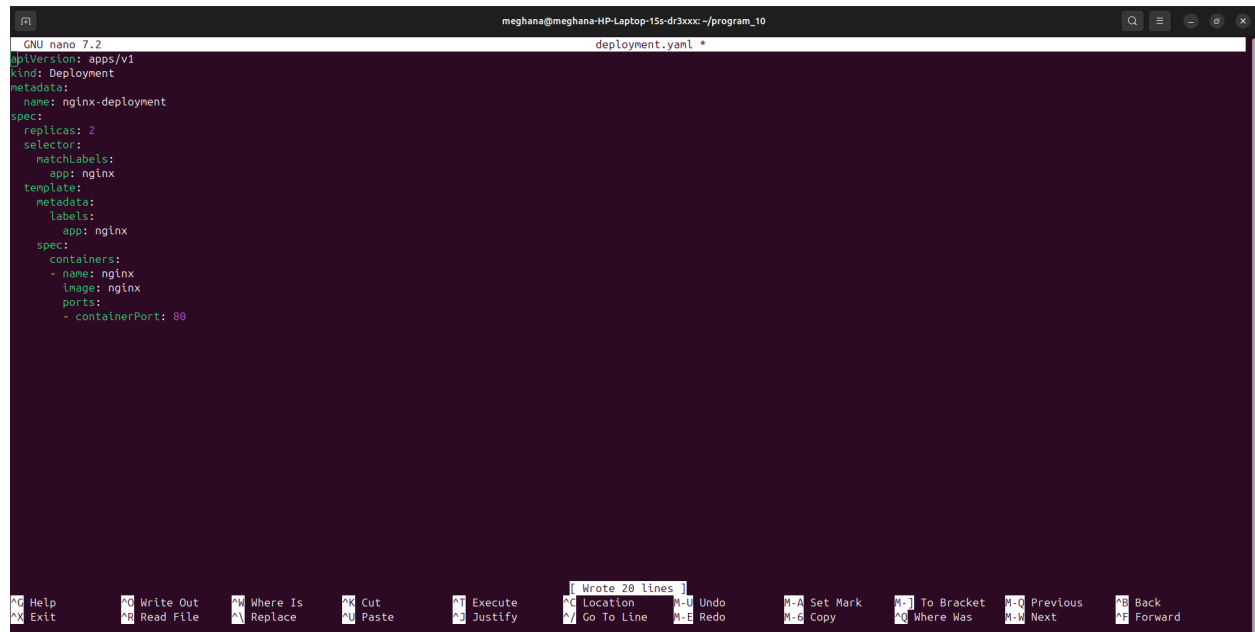
Step 2: Start Minikube

minikube start

A terminal window titled 'meghana@meghana-HP-Laptop-15s-dr3xxx: -' showing the execution of the 'minikube start' command. The output includes: 'minikube v1.37.0 on Ubuntu 24.04', 'Using the docker driver based on existing profile', 'Starting "minikube" primary control-plane node in "minikube" cluster', 'Pulling base image v0.0.48 ...', 'Restarting existing docker container for "minikube" ...', 'Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...', 'Verifying Kubernetes components...', a list of images being used (registry.k8s.io/metrics-server/metrics-server:v0.8.0, docker.io/kubernetes/dashboard:v2.7.0, gcr.io/k8s-minikube/storage-provisioner:v5, docker.io/kubernetes/metrics-scraper:v1.0.8), a note about dashboard features requiring the metrics-server addon, the command 'minikube addons enable metrics-server', and a final confirmation: 'Enabled addons: metrics-server, storage-provisioner, default-storageclass, dashboard' and 'Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default'. The prompt returns to 'meghana@meghana-HP-Laptop-15s-dr3xxx: \$'.

Step 3: Create deployment.yaml

nano deployment.yaml



The screenshot shows a terminal window with the nano text editor open. The file being edited is named 'deployment.yaml'. The content of the file is as follows:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

The nano editor's status bar at the bottom indicates 'Wrote 20 lines'.

Step 4: Create services.yaml

nano services.yaml



The screenshot shows a terminal window with the nano text editor open. The file being edited is named 'services.yaml'. The content of the file is as follows:

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

The nano editor's status bar at the bottom indicates 'Read 12 lines'.

Step 5: Apply Deployment and Service

kubectl apply -f deployment.yaml kubectl apply -f services.yaml

```
service/nginx-service unchanged
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment unchanged
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$ kubectl apply -f services.yaml
service/nginx-service unchanged
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$
```

Step 6: Get Pods, Deployment and Services

Get Pods:

kubectl get pods

Get Deployment:

kubectl get deployment

Get Services:

kubectl get services

```
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-app-787c9b7c57-nmffk          1/1     Running   8 (8h ago)  19d
demo-app-787c9b7c57-q7bz4          1/1     Running   8 (8h ago)  19d
my-first-pod                        1/1     Running   4 (8h ago)  11h
nginx-deployment-7cccd94f7-4w9pt    1/1     Running   0           2m57s
nginx-deployment-7cccd94f7-nt9fj    1/1     Running   0           8h
nginx-pod                           0/1     Completed 0           11h
nodejs-pod                         1/1     Running   8 (3m23s ago)  19d
program-devops-7-deployment-56bf5757cf-2qqqg  1/1     Running   3 (8h ago)  9h
program-devops-7-deployment-56bf5757cf-p9cqp  1/1     Running   3 (8h ago)  9h
program-devops-8-pod                1/1     Running   3 (3m23s ago)  9h
python-hello-7f4448c876-zqdz7       1/1     Running   10 (3m23s ago)  19d
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$ kubectl get deployment
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
demo-app  2/2     2             2           19d
nginx-deployment  2/2     2             2           5d7h
program-devops-7-deployment  2/2     2             2           9h
python-hello  1/1     1             1           40d
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$ kubectl get services
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes  ClusterIP   10.96.0.1     <none>         443/TCP          19d
nginx-service  NodePort    10.108.163.146 <none>         80:32452/TCP     5d7h
program-devops-7-service  NodePort    10.104.241.202 <none>         3000:30007/TCP   9h
program-devops-8-pod      NodePort    10.102.107.194 <none>         3000:32718/TCP   9h
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$
```

Step 7: Access Application using Minikube Service

minikube service nginx-service

Output:

```
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32452

```
Opening service default/nginx-service in default browser...
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_10$ [16343:16343:0105/220859.743163:ERROR:chrome/browser/ui/views/user_education/in
to show IPH IPH_DiscardRing before browser initialization complete; IPH will not be shown.
[16343:16343:0105/220859.877104:ERROR:chrome/browser/profiles/profile_attributes_storage.cc:1032] Failed to PNG encode the image.
```



Step 8: Launch Minikube Dashboard

minikube dashboard

